Choosing a Design: Which Assumption Will You Defend?

Experimentation, A/B Testing and Causal Inference (SDA-DSC-213), SDAIA Academy. Handout for Day 3, Hour 5 (slides 100, 117 and 120). All five Day 3 handouts use the same story: a tutoring programme at one school.

Step 1. The question to ask first

Every Day 3 method answers the same question, what did the treatment cause, under a different assumption. Choosing a design means choosing which assumption you are most willing to defend. Run the questions in this order (slides 120 and 193).

Ask	If yes	Assumption you are buying	Handout
Can you randomize?	run an experiment (Day 2)	only that the coin flips were done properly	Fisher, Neyman
Is there a threshold rule that assigns treatment?	regression discontinuity	units just above and just below the cutoff are alike; nobody can game the cutoff	5
Is there a nudge that shifts treatment but cannot touch the outcome directly?	instrumental variables	relevance, exclusion, independence; the answer is a LATE	4
Did treatment switch on at a known time for some units, with untreated comparisons?	difference-in-differences or synthetic control	parallel trends	3
Are the confounders rich and measured, with no plausible hidden one?	adjustment: g-formula, IPW, matching, doubly robust	unconfoundedness given X, plus overlap	1, 2
None of the above	say so	an honest "not identifiable" earns full marks in the capstone	

Step 2. The tutoring story, four ways

The same programme appeared under four designs across the handouts. Reading them side by side shows what each design needs.

Situation	Design	Estimate	Estimand	Weakest point

Weak students are sent to tutoring; prior grade is recorded	adjust for X	+9	ATE	an unrecorded reason for being sent (a parent's push)
Slots are allocated by lottery, attendance is voluntary	IV	+10	LATE for compliers	a lottery win that motivates study even without attending
A scholarship goes to scores of 70 and above	RDD	+0.30 GPA	effect at the cutoff	students who retake the test until they pass 70
One school opens a library, another does not	DiD	+6 points	ATT on the treated school	a shock that hit only School A in 2025

Step 3. What every observational estimate must carry

Slide 100 in one list. 1. The estimand and target population, stated first. 2. The adjustment set from the DAG, not from what improves fit. 3. An overlap check; trim to common support. 4. A balance check after adjustment (standardized mean differences below 0.1). 5. A doubly robust estimator with cross-fitting and a proper interval. 6. A sensitivity analysis for the unmeasured confounder (Day 4).

Step 4. Standard errors by design

Design	Where the randomness is	Standard error to use
Experiment, individual assignment	the coin flips	Neyman, or regression with HC2 or HC3
Experiment, group assignment	which groups were treated	clustered by group
Adjustment (IPW, AIPW)	sampling plus the fitted models	influence-function SE from the AIPW formula, or a bootstrap
DiD across regions	which regions were treated and when	clustered by region; needs enough clusters, roughly ten or more
IV	the instrument	2SLS standard errors; report the first-stage F
RDD	who lands near the cutoff	local-polynomial SE; show robustness to bandwidth

Step 5. The reconciliation habit (Lab 3)

When two designs are available, run both. Agreement between methods that rest on different assumptions is the strongest evidence observational data can give;

disagreement tells you which assumption is under strain. In Lab 3 the g-formula, IPW and AIPW agree near 4.5 points while the naive gap says 12 and the mediator-controlled regression says 2.2: the three that agree are the ones whose assumptions were met.

Three things to remember

1. Choose the design by the assumption you can defend, not by the method you like.

2. Name the estimand each design delivers; ATE, ATT, LATE and the effect at a cutoff are different quantities.

3. Match the standard error to where the randomness is.